

Personal significance is encoded automatically by the human brain: an event-related potential study with ringtones

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Abstract

In this human event-related brain potential (ERP) study, we have used one's personal – relative to another person's – ringtone presented in a two-deviant passive oddball paradigm to investigate the long-term memory effects of self-selected personal significance of a sound on the automatic deviance detection and involuntary attention system. Our findings extend the knowledge of long-term effects usually reported in group-approaches in the domains of speech, music and environmental sounds. In addition to the usual mismatch negativity (MMN) and P3a component elicited by deviants in contrast to standard stimuli, we observed a posterior ERP deflection directly following the MMN for the personally significant deviant only. This specific impact of personal significance started around 200 ms after sound onset and involved neural generators that were different from the mere physical deviance detection mechanism. Whereas the early part of the P3a component was unaffected by personal significance, the late P3a was enhanced for the ERPs to the personal significant deviant suggesting that this stimulus was more powerful in attracting attention involuntarily. Following the involuntary attention switch, the personally significant stimulus elicited a widely-distributed negative deflection, probably reflecting further analysis of the significant sound involving evaluation of relevance or reorienting to the primary task. Our data show, that the personal significance of mobile phone and text message technology, which have developed as a major medium of communication in our modern world, prompts the formation of individual memory representations, which affect the processing of sounds that are not in the focus of attention.

Introduction

Fast and efficient detection of changes in the environment is important for adaptive behaviour. The human auditory modality reveals a highly developed system that continuously models the acoustic environment based on auditory sensory memory representations even outside the focus of attention and automatically registers changes that do not fit the model. This monitoring may result in the elicitation of an involuntary attention switch prioritizing the processing of the change and initiating an orienting response (Sokolov, 1963; Öhman, 1979; Näätänen, 1992). By comparing event-related potentials (ERPs) elicited by rarely presented deviant tones with those elicited by frequently presented standard tones in experimental passive oddball paradigms, one can probe the automatic deviance detection and involuntary attention system (e.g. Schröger, 1998). The non-volitional detection of such deviance is indicated by the mismatch negativity (MMN) component (Näätänen *et al.*, 1978; Schröger, 1997; Näätänen & Winkler, 1999; Picton *et al.*, 2000), which is measurable as a negative deflection maximal over frontocentral recording sites with reversed polarity over posterolateral recording sites (when nose reference is used) that peaks between 100 and 250 ms. If the deviant is of sufficient saliency (Yago *et al.*, 2001), the MMN is followed by a frontocentrally distributed positive deflection, the P3a, an ERP

component reflecting the involuntary orienting of attention towards the perturbing sound (e.g. Squires *et al.*, 1975; Escera *et al.*, 2000; Friedman *et al.*, 2001).

Several studies reported that the preattentive processing of a change is modulated by long-term memory representations (for an overview see Näätänen *et al.*, 2001; Schröger *et al.*, 2004), most frequently investigated in the domains of speech and music. In detail, using speech stimuli, material we are most familiar with, it has been shown robustly that enduring long-term memory representations concerning phonemes, syllables or words of the own language affect the preattentive auditory processing of an acoustic change (Dehaene-Lambertz, 1997; Näätänen *et al.*, 1997; Cheour *et al.*, 1998; Winkler *et al.*, 1999; Pulvermüller *et al.*, 2001, 2004; Jacobsen *et al.*, 2004; reviewed in Näätänen, 2001). In addition to language specific effects, the preattentive processing is influenced by the familiarity of a voice, which can carry important information about the personal relevance of the perceived speech sound (Beauchemin *et al.*, 2006; Holeckova *et al.*, 2006). In the domain of music, long-term training leads to improved extraction of regularities in an auditory stimulation and to improved detection of smaller acoustic changes (e.g. Koelsch *et al.*, 1999; Tervaniemi *et al.*, 2001; Nager *et al.*, 2003; Fujioka *et al.*, 2004; van Zuijen *et al.*, 2004). There is some research showing long-term memory effects using environmental sounds (Escera *et al.*, 2003; Frangos *et al.*, 2005; Jacobsen *et al.*, 2005; Hauk *et al.*, 2006). As most of these studies chose group approaches [for an exception see Escera *et al.* (2003) who computed ERPs for identifiable sounds, and

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non-identifiable ones, respectively, on the basis of individual participants' identification ratings], electrophysiological research investigating the influence of personal significance is rare. Only a few behavioural studies, mostly using the dichotic listening paradigm, suggest that personal significance of acoustic sounds is processed even outside the focus of attention (Formby, 1967; Moray, 1959; Wood & Cowan, 1995).

The present study tested whether the long-term memory representation of personal significance of one's own ringtone affects the automatic deviance detection and involuntary attention system when it occurs among other environmental sounds sharing the same meaning at a semantic level but lacking personal significance.

Materials and methods

Participants

Fifteen volunteers participated in the study (eight men and seven women). Their median age was 23 years (range 19–29). The 16 originally scheduled participants were grouped into pairs to receive physically identical stimulation (see experimental design below). Participants reported normal auditory and normal or corrected-to-normal visual acuity and no neurological, psychiatric, or other medical problems and were treated according to the ethical principles of the World Medical Association (1996; Declaration of Helsinki). They gave informed consent, and received course credits or monetary compensation. Participants needed to display an exogenous N1 component in their ERP waveform, based on the data gathered in the experimental session, to be part of the analysed experimental group, as it reflects normal auditory system functioning. One participant and therefore also its paired associate were excluded from further analysis because of this criterion. The sixteenth scheduled participant cancelled, and therefore its paired associate was also excluded. As a consequence, data of 12 participants were analysed eventually.

Stimuli and experimental design

The personal ringtones to incoming text messages of the 16 scheduled participants were used as stimuli. These were all different, were recorded prior to the experimental session (HEADacoustics HMS III.0, HEADacoustics GmbH; Herzogenrath, Germany), were normalized in intensity, and had an average duration of 964 ms (range 445 ms to 1699 ms). Counterbalanced across participants, each stimulus equally served as a frequently presented standard, as personally significant deviant and as personally non-significant deviant.

The design had the following constraints; (i) all pairs of participants received physically identical stimulation with their own and the other person's ringtone; (ii) each standard served as a non-significant deviant ringtone (for one member of a pair) and as a personally significant deviant ringtone (for the other member of that pair), and (iii) each deviant was also used as a standard. These constraints, which also required that two different stimuli served as standards for each participant, resulted in the yoked design presented in Table 1. Four different stimuli were used for each participant (personally significant deviant, non-significant deviant, two non-significant standards), and participants were grouped accordingly so that the resulting quadruples of stimuli were physically different and clearly distinguishable from each other from sound onset.

After the experimental session, all participants rated the ringtones they were presented with according to familiarity (1, highly familiar to

TABLE 1. Yoked experimental design

Participant	First part		Second part	
	Standard stimulus (85%)	Deviant stimuli (each 7.5%)	Standard stimulus (85%)	Deviant stimuli (each 7.5%)
1	02	01/03	04	01/03
2	03	02/04	01	02/04
3	04	03/01	02	03/01
4	01	04/02	03	04/02
5	06	05/07	08	05/07
6	07	06/08	05	06/08
7	08	07/05	06	07/05
8	05	08/06	07	08/06
9	10	09/11	12	09/11
10	11	10/12	09	10/12
11	12	11/09	10	11/09
12	09	12/10	11	12/10
13	14	13/15	16	13/15
14	15	14/16	13	14/16
15	16	15/13	14	15/13
16	13	16/14	15	16/14

Numbers indicate the 16 scheduled participants and their personal ringtones. Yoked into pairs (i.e. Participant 1 and 3, 2 and 4, and so forth), participants received physically identical stimulation with the role of personally significant and non-significant ringtone exchanged. The respective personal ringtone is printed in boldface. 1st part and 2nd part refer to the blocks which included different standard stimuli. Data of two participants (1 and 11) could not be included in data analysis and thus the data of the respective paired associates (3 and 9) as well.

5, unfamiliar/unknown). The stimuli presented as non-significant were rated 3.4 (mean), whereas the stimuli presented as the significant ones were rated 1.1 (mean), resulting in a significant difference ($t_{11} = 8.7$, $P < 0.01$). Participants were asked to specify the frequency of receiving a message and the time which had elapsed since they had chosen this ringtone. On average, participants reported to receive three text messages per day, ranging from one to ten, and that they had been using the particular ringtone for 15 months (mean, ranging from 3 to 72 months).

Procedure

The participants were comfortably seated in an electrically shielded and sound-attenuated experimental chamber (Industrial Acoustics Company GmbH, Niederkrüchten, Germany) and were instructed to watch a self-selected, subtitled, and silenced video while ignoring the auditory stimulation. Oddball stimulus sequences were composed of a ringtone, which served as the standard (85% of the trials) and two ringtones, which served as deviants (7.5% of the trials each). This two-deviant paradigm was preferred over two separate sequences with one deviant each, as less predictability of the occurring deviant is likely to improve deviance-related effects.

The respective standard and the two deviants were presented in pseudorandomized order while at least two standards were presented between successive deviants. A total of 1000 trials (850 standards, 75 personally significant deviants, 75 non-significant deviants) were presented in five stimulus blocks. Another 1000 trials were presented with a second standard. Stimuli were presented binaurally via headphones (Sennheiser HD 25; Sennheiser electronic GmbH & Co. KG, Wedemark, Germany) with a stimulus onset-to-onset interval of 1800 ms at 65 dB SPL intensity. After the experimental session, participants reported that they perceived the own ringtone as their own and the other tones as other persons' ringtones.

Electrophysiological recordings

The electroencephalogram (EEG; Ag/AgCl electrodes, EASYCAP GmbH, Herrsching-Breitbrunn, Germany; BrainAmp EEG amplifier; BrainAmp Products GmbH, Gilching, Germany) was recorded continuously from 28 standard scalp locations according to the extended 10–20 system (FP1, FPz, FP2, F7, F3, Fz, F4, F8, FC5, FC1, FC2, FC6, T7, C3, Cz, C4, T8, CP1, CP2, P7, P3, Pz, P4, P8, O1, O2, LM, RM) with a sampling rate of 500 Hz and 16-bit resolution. The reference electrode was placed on the tip of the nose, the ground electrode at AFz. Electroocular activity (EOG) was recorded with two bipolar electrode pairs, the vertical EOG from the right eye by one supraorbital and one infraorbital electrode, and the horizontal EOG from electrodes placed lateral to the outer canthi of the two eyes. Impedances were kept below 5 kOhm. On-line filtering was carried out using a 100-Hz low-pass and a 0.05-Hz high-pass filter.

Data analysis

EEG recordings were off-line filtered band-pass with a finite impulse response filter; 1601 points, critical frequencies of 1 Hz (high-pass) and 15 Hz (low-pass). EEG epochs of 1100 ms length, including a 100-ms pre-stimulus baseline, which did not show an amplitude change exceeding 100 μ V at any of the recording channels, were averaged separately for each condition (standard, deviant, personally significant deviant, non-significant deviant) and participant (eeprobe 3.2; ANT, Enschede, Netherlands). Grand-averages were subsequently computed from the individual-subject averages. As a result of the experimental design, each stimulus contributed equally to each grand-average response and difference wave, thus eliminating the effects of physical stimulus features from the grand-averages. For ERP quantification, amplitudes were measured as the mean voltage in a time window centred on the respective grand-average difference wave latency of maximal voltage deflection. For the deviance-related effects, difference waves were calculated by subtracting the standard ERPs from the deviant ERPs. The effects related to personal significance were revealed in the difference waves 'personally significant deviant-minus-standard' minus 'non-significant deviant-minus-standard'.

Statistical analysis

Deviance-related effects, the presence of MMN and P3a, were tested at electrodes of maximal voltage deflections of the difference waves, which corresponded to positions of maximal voltage deflections usually revealed in former studies. To quantify the MMN, average voltage values for the standard and deviant ERPs at electrode Fz were analysed in a time window of ± 20 around the peak latency of 132 ms (Schröger, 1998). The same statistical procedure was applied for P3a in the time window around the peak at $256 \text{ ms} \pm 40$ for Cz data (Escera *et al.*, 2000). Two-tailed dependent *t*-tests were also used to check these deviance-related components for possible effects of personal significance of the deviant stimulus.

Effects related to personal significance were analysed statistically via three-way repeated-measures ANOVAs [personal significance (yes, no), anterior-posterior (frontal F7, F3, Fz, F4, F8; central T7, C3, Cz, C4, T8; parietal P7, P3, Pz, P4, P8), lateral scalp location (7-, 3-, z-, 4-, & 8-lines)] in the time windows $204 \text{ ms} \pm 20$, $396 \text{ ms} \pm 20$ and $704 \text{ ms} \pm 80$.

In order to test topographical differences between deviance-related effects (MMN, P3a) and the subsequent effects of personal signifi-

cance ($204 \text{ ms} \pm 20$, $396 \text{ ms} \pm 20$), ERP difference wave data were analysed in two three-way repeated measures ANOVAs [condition (deviance-related effect vs. effect related to personal significance), anterior-posterior, lateral scalp location]. Data were vector-normalized for individual participants in order to remove global amplitude differences, following the suggestions of McCarthy & Wood (1985).

For effects with more than one degree of freedom, the original degrees of freedom are reported along with the corrected probability (Greenhouse-Geisser).

Results

Deviance-related effects

The ERPs elicited by the deviants (Fig. 1A), revealed an enlarged negativity compared to the ERPs elicited by the standards, in the time range between 112 and 152 ms relative to stimulus onset (MMN; peak at Fz 132 ms, $-3.4 \mu\text{V}$; $t_{11} = 10.15$, $P < 0.01$). The MMN had an anterior distribution and showed polarity inversion at the mastoids (Fig. 1B left column). Subsequently, deviants elicited a positive frontocentrally distributed deflection (P3a; Fig. 1B right column) between 216 and 296 ms (peak at Cz 256 ms, $3.9 \mu\text{V}$; $t_{11} = -4.73$, $P < 0.01$). Neither the MMN nor the P3a amplitudes differed between personally significant deviant and non-significant deviant ($P > 0.1$).

Effects related to personal significance

An influence of personal significance was assessed by comparing the deviance-related effects between the personally significant and the non-significant deviants (Fig. 2A). This comparison showed effects in three time ranges. First, the personal ringtone evoked an increased negative potential between 184 and 224 ms at several posterior electrodes (Fig. 2B left column) as indicated by the significant interaction of personal significance $\times$ anterior-posterior (peak at Pz 208 ms, $-2.9 \mu\text{V}$; $F_{2,22} = 11.23$, $P < 0.01$). Besides this interaction, neither the main effect of personal significance, nor any other interaction involving this factor reached significance in this time window ($P > 0.05$). Second, the personal ringtone elicited an increased positivity between 376 and 416 ms at frontal electrodes (Fig. 2B middle column; peak at Fz 398 ms, $-2.1 \mu\text{V}$, interaction of personal significance $\times$ anterior-posterior $F_{2,22} = 5.61$, $P < 0.05$). There was no other significant interaction involving the factor personal significance in that time window ($P > 0.1$). Third, the ERPs to the personal ringtone revealed a broadly distributed negative deflection between 624 and 784 ms, whereas the ERPs to the non-significant ringtone did not (Fig. 2B right column; main effect condition $F_{1,11} = 14.58$, $P < 0.01$). An additional interaction of personal significance $\times$ lateral scalp location ($F_{4,44} = 5.16$, $P < 0.02$) showed after subsequent *posthoc* testing a significantly diminished effect at the very left lateral electrode positions. There was no other significant interaction involving the factor personal significance in that time window 624–784 ms ($P > 0.1$).

Topographical differences

Analysis of topographical differences between the mere MMN (112–152 ms) and the first modulation of this deviant vs. standard contrast by personal significance (184–224 ms) revealed a significant interaction of condition $\times$ anterior-posterior ($F_{2,22} = 84.91$,

Deviance-related effects

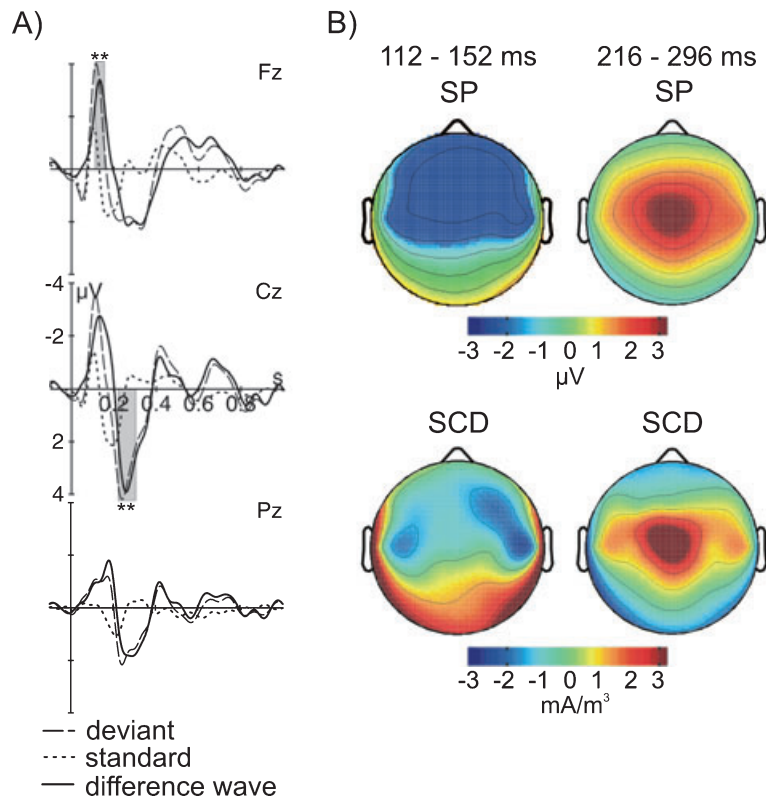


FIG. 1. Deviance-related effects. (A) Grand-average ERPs at representative frontocentral (Fz), central (Cz) and parietocentral (Pz) electrodes. Shown are ERPs to the standards (dashed), ERPs to the deviants (personally significant and non-significant averaged together, broken) and difference waves (deviant minus standard, solid). The bars and the asterisks mark the statistically analysed time windows and positions of significant differences between ERPs ($**P < 0.01$). (B) Mean scalp potential (SP) and scalp current density distributions (SCD) for deviance-related effects (112–152 ms, MMN; 216–296, P3a).

Effects related to personal significance

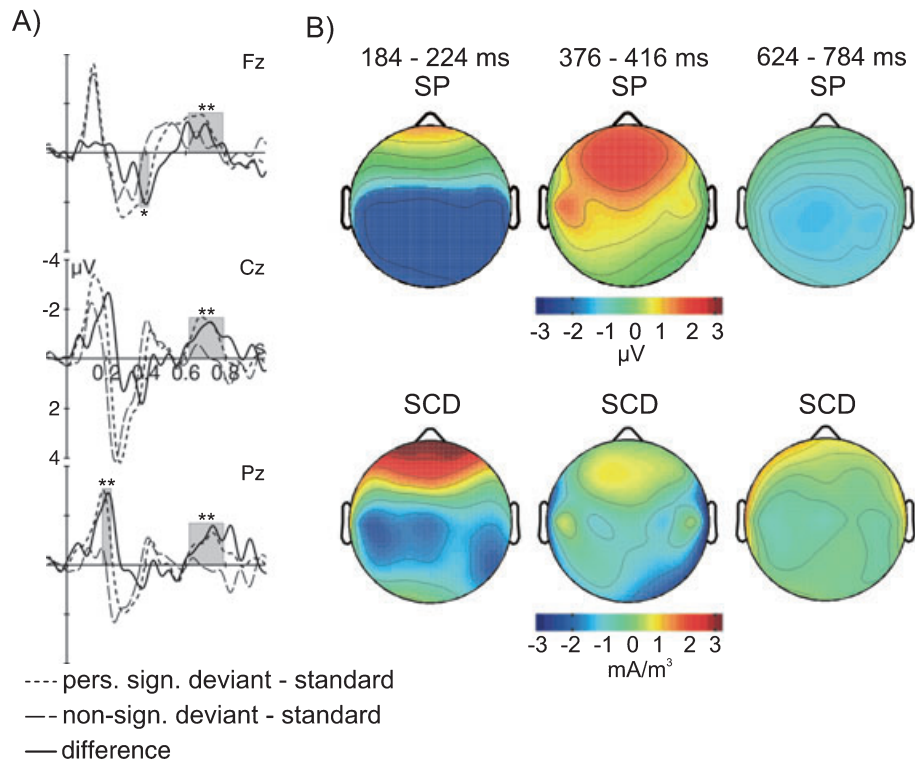


FIG. 2. Effects related to personal significance. (A) Grand-average difference waves at representative frontocentral (Fz), central (Cz) and parietocentral (Pz) electrodes. Shown are difference waves 'personally significant deviant-minus-standard' (dashed), 'non-significant deviant-minus-standard' (broken) and their difference (solid). The bars and the asterisks mark the statistically analysed time windows and positions of significant differences between ERPs ($**P < 0.01$; $*P < 0.05$). (B) Mean scalp potential (SP) and scalp current density distributions (SCD) for effects related to personal significance (184–224 ms; 376–416 ms; 624–784 ms).

$P < 0.01$), which was due to a frontally distributed MMN and a posterior effect related to personal significance. No other interaction involving the factor condition reached significance ($P > 0.05$).

The second effect related to personal significance occurred at the terminal part of the P3a in the time window 376–416 ms. Indicated by a significant interaction of condition $\times$ lateral scalp location ($F_{4,44} = 16.05$, $P < 0.01$) and of condition $\times$ anterior-posterior $\times$ lateral scalp location ($F_{8,88} = 3.25$, $P < 0.05$), these two effects had a different topographical distribution (centrally distributed deviance-related effect, not lateralized; frontally distributed effect related to personal significance, lateralized to the left).

Discussion

The purpose of the present study was to investigate the role of individually significant long-term memory representations on the automatic deviance detection and involuntary attention system. Personal ringtones to incoming text messages, gaining increasing importance in the world of electronic communication, served as stimuli. In every-day life, these often have to be identified against concurrent ringtones of others. Thus, it is necessary to store the acoustic pattern of that stimulus and to memorize the association to personal significance.

A two-deviant passive oddball paradigm was used in the present experiment, in which each rarely presented deviant (personally significant vs. non-significant) occurred with the same probability. All stimuli, being ringtones, shared the same meaning for each participant, but only one was of personal significance. Brain responses to the personally significant auditory stimulus, i.e. the personal ringtone, were compared with the responses to a non-significant auditory stimulus, i.e. ringtone of another person.

The comparison of the ERPs elicited by deviants with those elicited by standards, which were ringtones as well, revealed the predicted MMN and P3a components. Thus, both rarely presented deviants were detected automatically by the human brain and triggered an involuntary switch of attention (e.g. Näätänen, 1992; Escera *et al.*, 2000). The used yoked design included the same physical contrast between standard and personally significant and non-significant deviant stimuli over all participants. Hence, ERP results obtained for the contrast of personally significant deviant vs. non-significant deviant could only be due to the influence of individual long-term memory representations on the automatic deviance detection and involuntary attention system. The first effect, which was solely related to personal significance, occurred relatively early, directly following the MMN. From a chronometric point of view, this suggests that the information about personal significance was extracted after the information of physical deviance (MMN), but before the initiation of an attention switch (P3a). MMN and the early effect related to personal significance were characterized by different topographical distributions with frontal maxima of the MMN and a posterior localized effect related to personal significance. Thus, spatial configurations of underlying generators, as also reflected by the scalp current densities, differ at least partly. In turn, the early effect related to personal significance is not a mere modulation of the late part of the MMN, but reflects an additional process that appears to require auditory deviance detection as a prerequisite. The former can be speculated to result from the outcome of successful deviance detection, and further processes this information with respect to personal significance. We suggest the further processing can be regarded as a result of enhanced elaboration of the own ringtone and thus as a result of broader network activity (e.g. Pulvermüller, 1999). Widespread neurons associated as a functional unit when detecting the personally significant sound could

lead to these differences in activation following deviance detection. Spatial integration of the signal, possible elicitation of an orienting response and release of a motor disposition to act might be speculative implications. Alternatively, this negativity might also be a subcomponent of the N2, namely the N2c. This subcomponent has been suggested to reflect a process related to stimulus classification while the N2a (MMN) and N2b reflect different stages of mismatch detection (Pritchard *et al.*, 1991). In other words, this negativity might indicate that the ringtone was categorized as one's own.

Previous studies, having investigated long-term memory effects of familiarity in passive oddball paradigms (Frangos *et al.*, 2005; Jacobsen *et al.*, 2005; Beauchemin *et al.*, 2006; Hauk *et al.*, 2006) reported modulations of MMN amplitude, which was not the case for the variation of personal significance in our study. In those previous studies, familiarity was varied in different ways: the contrast familiar vs. unfamiliar corresponded to known vs. unknown (Beauchemin *et al.*, 2006; Hauk *et al.*, 2006), and meaningful vs. meaningless stimuli, respectively (Frangos *et al.*, 2005; Jacobsen *et al.*, 2005). In the present study, however, all stimuli were identifiable as signal stimuli, thus shared the same meaning on the group-level for all participants. The absence of an MMN modulation appears to be due to the fact that we did not vary familiarity in the sense of a known vs. unknown or meaningful vs. meaningless contrast. The manipulation of personal significance did not affect the MMN system, but brought into play additional neural processes subsequent to the MMN.

Moreover, in most of the previous studies the influence of familiarity on MMN amplitude was confounded with the physical stimulus characteristics obscuring an unambiguous interpretation of results. The present study used a yoked design with large variation in physical stimulus characteristics that avoids this confound. Nevertheless, although this individual approach allows analysis of personal long-term memory representation, it increases variation between participants in the ERP data, which could in principle obscure additional small ERP modulations.

Although not formally tested, the present results suggest a replication of reports of a two-phasic P3a-component (e.g. Escera *et al.*, 1998, 2001; Debener *et al.*, 2005). The early P3a subcomponent, characterized by a centrally dominant distribution was elicited for both deviants equally, whereas the late P3a was modulated by personal significance. It has been shown that the early P3a seems to be more associated with the violation of the regularity registered by the automatic deviance detection system (Escera *et al.*, 1998, 2001) compared to the late P3a, which reflects the actual orienting of attention and was enhanced for identifiable and familiar sounds in previous studies as well (e.g. Escera *et al.*, 2003; Holeckova *et al.*, 2006).

Escera *et al.* (2003) reported that the modulation of the late P3a component in their study was sensitive to attentional manipulations (see also Mecklinger *et al.*, 1997; Opitz *et al.*, 1999). In contrast to our study, they used neutral environmental sounds presented as different kinds of novels, which could be known (identifiable) or unknown (non-identifiable), among non-environmental sounds presented as standards. In this context, it would be interesting to know whether also the present effect (i.e. the enhancement in late P3a amplitude due to personal significance) is sensitive to manipulations of attentional demands.

Subsequent to the MMN-P3a-complex, there was a widely distributed negative deflection, which appeared to be solely due to the processing of the personally significant deviant. As it was not elicited for the non-significant deviant, it might reflect further analysis of the personal ringtone involving evaluation of relevance in the current situation. Such an evaluation seems reasonable as in normal situations,

the ringing of the own mobile phone asks for some behaviour (e.g. looking for the mobile, fetching it, pressing a button, and the like) which is not the case when another person's mobile phone rings. This late brain wave may also indicate reorienting to the primary task (i.e. watching the movie) which becomes necessary, because the personally significant sound elicits a more distinct involuntary attention switch. Often, larger distraction as indicated by P3a and behavioural impairment is associated with an enlarged re-orienting negativity (RON; Yago *et al.*, 2001; Berti *et al.*, 2004). Moreover, as the scalp potential and scalp current source distributions argue against a generation of this effect in primary auditory areas (Fig. 2B), it appears that this effect occurs at some post-perceptual stage of processing.

In summary, the present findings extend our knowledge about the influence of long-term memory representations on auditory processing via the automatic monitoring of the acoustic environment. Especially, personal significance of a stimulus prompts the formation of individual memory representations that do not only modulate involuntary attention processes, but also activate different neural generators before an attention switch is initiated and leads to qualitatively different and additional subsequent processing of the personally significant sound.

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Abbreviations

EOG, electrooculogram; ERP, event-related potential; MMN, mismatch negativity.

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